

EXECUTIVE SUMMARY

The logistics industry plays a vital role in ensuring the efficient movement of goods from suppliers to customers. With the rapid growth of e-commerce and global supply chains, timely delivery has become one of the most important factors influencing customer satisfaction and business success. Delayed deliveries can result in increased operational costs, customer dissatisfaction, and reduced organizational efficiency. Therefore, predicting the likelihood of late deliveries before they occur has become an important area of research in logistics and supply chain management.

This research presents a machine learning-based solution for predicting late delivery risk using the **APL Logistics dataset**. The project was completed as part of a **three-month internship at Unified Mentor** under the Machine Learning domain. The objective was to develop an intelligent prediction system capable of identifying shipments that are at risk of delayed delivery, enabling organizations to make proactive operational decisions.

The project began with data collection and exploratory data analysis to understand the characteristics of the logistics dataset. Various preprocessing techniques were then applied, including handling missing values, removing duplicate records, encoding categorical variables using Label Encoding, and standardizing numerical features using StandardScaler. In addition, several business-oriented features such as Shipping Pressure Index, Customer Value Score, Profit Margin, Shipping Efficiency, Order Complexity Score, and Market Risk Score were engineered to improve the predictive capability of the model.

Following data preprocessing, an **XGBoost Classifier** was selected as the machine learning algorithm due to its high accuracy, efficiency, and ability to handle structured data effectively. The processed dataset was divided into training and testing sets, and the model was trained to classify whether an order was likely to experience delayed delivery. Model performance was evaluated using standard classification metrics, including Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix, demonstrating the effectiveness of the proposed approach.

To enhance the practical usability of the system, an interactive dashboard was developed using the **Streamlit** framework. The dashboard integrates the trained machine learning model with interactive visualizations and consists of multiple modules, including the Executive Dashboard, Risk Analytics, Regional Analytics, Prediction Center, Model Performance, and About Project. These modules enable users to explore logistics data, analyze business trends, and generate real-time predictions through a simple graphical interface without requiring programming knowledge.

The developed system demonstrates how machine learning and data visualization can be integrated to support predictive decision-making in logistics operations. By identifying high-risk shipments in advance, organizations can optimize delivery planning, improve resource allocation, reduce operational risks, and enhance customer satisfaction.

Overall, the project successfully achieved its objectives and illustrates the practical application of predictive analytics in logistics management. The proposed solution provides a scalable framework that can be further enhanced by incorporating real-time traffic data, weather information, GPS tracking, and additional machine learning techniques in future research. It highlights the growing importance of artificial intelligence and data-driven technologies in modern supply chain and logistics operations.